TEXAS STAAR

GRADE 8

Reading Language Arts & Mathematics

All-in-One Study & Practice Guide

8 Full-Length Practice Tests · Complete Skill Review · Deep Answer Explanations

Short & Extended Constructed Response Mastery · Score Diagnostics

100% aligned to the Texas Essential Knowledge and Skills (TEKS)

2027 EDITION

Adrian Ross

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All passages, questions, graphics, and practice materials in this book are **original works** created for instructional practice. They are designed to reflect the style, format, difficulty, and standards coverage of the STAAR Grade 8 assessment; they are **not** actual test questions and are **not** reproduced from any secure or released state testing material.

About This Edition

This 2027 Edition reflects the STAAR program **as of November 5, 2026**, including the online test format and the current question types, scoring, and TEKS blueprint. Under Texas House Bill 8 (2025), **spring 2027 is the final STAAR administration** for grades 3–8; a new assessment, the Student Success Tool, is scheduled to replace STAAR beginning in the 2027–2028 school year. This book is built to prepare students thoroughly for that final STAAR year.

While every effort has been made to ensure accuracy, test specifications can change. Families and educators should confirm current details on the official Texas Education Agency website.

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How this book is organized: First you *learn* each tested skill, then you *practice* it in short drills, then you *simulate* the real test with eight full-length practice tests, and finally you *diagnose* your results to see exactly what to study next.

Welcome to Your STAAR Grade 8 Study Guide

Dear Student,

The STAAR Grade 8 tests in Reading Language Arts and Mathematics can feel like a lot — two subjects, many skills, a timed online screen, and even written essays. This book exists to make all of that feel familiar long before test day.

Here is the promise of this book: **everything inside is on the test, and nothing on the test is left out.** Every lesson, every practice question, and every writing prompt is tied directly to the Texas Essential Knowledge and Skills (TEKS) that STAAR measures at Grade 8. You will not waste a single minute on material that will not help you.

You will move through four stages:

1. **Learn** — short, clear lessons that teach every tested skill.
2. **Practice** — quick drills so each skill sticks.
3. **Simulate** — eight full-length practice tests that look and feel like the real thing.
4. **Diagnose** — answer explanations and charts that show you exactly what to review next.

Work at a steady pace, check every answer explanation (even for questions you got right), and use the diagnostic charts to spend your study time where it matters most. Confidence on test day comes from practice, and this book gives you plenty.

You can do this. Let's get started.

— *Adrian Ross*

A note for parents and teachers: A full guide for supporting a student with this book — including how to read the diagnostic charts and how to coach the written responses — begins on the "A Guide for Parents and Teachers" page near the back.

How to Use This Book

This book works for three kinds of readers. Find the path that fits you.

Path A — The Student Studying on Your Own

1. **Start with "Getting Started."** Read how the test works, how it is scored, and how the online question types look on paper. About 30 minutes — and it saves you from surprises.
2. **Pick a study calendar.** Use the 4-Week Sprint if the test is close, or the 8-Week Plan if you have more time.
3. **Learn, then practice.** Read the skill lessons and do the warm-up drills before a full test.
4. **Take one full test at a time, under test-like conditions** — quiet room, no phone, timed.
5. **Review every explanation,** including for questions you answered correctly.
6. **Fill in your diagnostic chart** after each test, then study your weakest strand next.

Path B — The Parent or Tutor Helping a Student

- Use the **"A Guide for Parents and Teachers"** section at the back first.
- Sit with the student for the **first** practice test to model pacing, then let them work more independently.
- Focus praise on **effort and strategy**, not just the score. The tracking sheets make growth visible.

Path C — The Classroom Teacher

- Lessons and drills map one-to-one to the **TEKS**; the **TEKS Coverage Index** lists where every standard is taught and practiced.
- Each full-length test is blueprint-matched — usable as a benchmark or mock administration.
- Explanations include the TEKS code and *why each wrong choice is wrong*, useful for reteaching.

The Symbols You Will See

Symbol	Meaning
★	**Readiness standard** — tested heavily every year. Master these first.
○	**Supporting standard** — tested less often, but still fair game.

Key Idea	The one thing to remember from a lesson.
Watch Out	A common Grade 8 mistake to avoid.
Try It	A quick practice item to check yourself.
Strategy	A test-taking tip tied to the question in front of you.

Tip: Keep a pencil, scratch paper, and the Grade 8 Mathematics Reference Sheet nearby while you work. On the real test these tools are available to you — practice using them now.

Inside the STAAR Grade 8 Test

The Final STAAR Year

STAAR — the State of Texas Assessments of Academic Readiness — measures how well students have learned the **Texas Essential Knowledge and Skills (TEKS)** for their grade. Under Texas House Bill 8 (2025), **spring 2027 is the last STAAR administration** for grades 3–8. A new assessment called the Student Success Tool is scheduled to replace it in 2027–2028. This book prepares you fully for that final STAAR year.

A Computer-Based Test

STAAR is given **online**, on a computer. The real test includes question types beyond simple multiple choice — you might type an answer, choose from a drop-down menu, drag items into place, or build an equation. A paper book can't make you click and drag, so this book does the next best thing: it **shows every online question type on paper** and teaches you exactly how to answer it. The *thinking* is identical; only the clicking is different.

What Grade 8 Students Take

At Grade 8, students are assessed in **Reading Language Arts, Mathematics, Science, and Social Studies.** This book focuses on the two taken by every Grade 8 student every year: **Reading Language Arts and Mathematics.**

The Mathematics Test at a Glance

Feature	Grade 8 Mathematics
Total questions	**40**
Total points	**48** (32 one-point + 8 two-point questions)
Reporting categories (strands)	4 — Numerical; Computations & Algebraic; Geometry & Measurement; Data & Financial Literacy
Question types	Multiple choice, multiselect, inline choice, text entry/griddable, drag-and-drop, equation editor, graphing
Tools provided	**Calculator** and the **Grade 8 Mathematics Reference Sheet**

The Reading Language Arts Test at a Glance

Feature	Grade 8 Reading Language Arts
Total questions	**~45**
Total points	**56**
Reading section	~28 questions / ~30 points — includes **2 Short Constructed Responses** (2 pts each)
Writing section (Revising & Editing)	~17 questions / ~26 points — includes **1 Extended Constructed Response** (10 pts)
Passages	Literary and informational; single passages and a paired set
Tools provided	Online **dictionary/thesaurus**

How Much Time?

STAAR for grades 3–8 is given with a generous time window of **up to four hours**. It is **not a speed test** — most students finish comfortably. Still, practicing with a timer builds a steady, calm pace.

> **Watch Out:** Written responses carry real points — up to **14 of the 56 RLA points** come from the SCRs and ECR. Never leave them blank. A partial, organized response almost always earns more than nothing.

How STAAR Is Scored

From Raw Points to a Performance Level

Each question earns **raw points** (1 or 2 for most; the Extended Constructed Response is worth 10). Your total raw points convert to a **scale score**, which places you in one of four **performance levels**:

Performance Level	What It Means
Masters Grade Level	Strong command of the grade's content; ready for the next grade with little support.
Meets Grade Level	Solid understanding; likely to succeed next grade with some support.
Approaches Grade Level	Some understanding; likely needs targeted support next grade.
Did Not Meet Grade Level	Not enough understanding yet; needs significant support.

The goal for most students is **Meets Grade Level** or higher — and this book is built to get you there.

How Written Responses Are Scored

- **Short Constructed Response (SCR)** — a few sentences answering a specific question, scored **0 to 2 points**. Earn full credit by making a clear point and backing it with correct **evidence from the text**.
- **Extended Constructed Response (ECR)** — a full multi-paragraph essay, scored **0 to 10 points**. The rubric rewards a clear central idea or claim, organized paragraphs, specific supporting evidence, and correct language and conventions.

You will learn to earn these points step by step in the **SCR Mastery** and **ECR Mastery** sections, which include real-style prompts and sample responses scored at high, middle, and low levels.

Tracking Your Own Scores

After each practice test, use the **Score-Tracking Sheets** and **Strand Diagnostic Charts** at the back. They turn raw points into a performance estimate and show you **which strand or skill to study next.** Watching your scores climb from Test 1 to Test 4 is the clearest sign your work is paying off.

Item-Type Field Guide

How Online Questions Look on Paper

STAAR is taken on a computer, so some questions ask you to do more than bubble a letter. This guide shows **every question type you may meet**, how we present it on paper in this book, and a quick example. The skill tested is always a Grade 8 TEKS skill — only the way you record your answer changes.

> **Key Idea:** On screen you *click, type, or drag.* On paper you *circle, write, or draw an arrow.* The thinking is exactly the same.

1. Multiple Choice

Online: Click one choice (A–D). **On paper:** Circle the letter of the one best answer.

Example (Math). Which number is the greatest? A. 0.6 B. ⅗ C. 0.58 D. ½ *(Answer: A)*

2. Multiselect

Online: Check every correct box. **On paper:** Circle **all** correct letters; the question says how many.

Example (Math). Select the **two** numbers that are irrational. A. √49 B. √10 C. π D. 0.25 *(Answers: B and C)*

3. Inline Choice (Drop-Down)

Online: Click a drop-down inside the sentence. **On paper:** Each drop-down is a box of choices; circle the correct one.

Example (Revising & Editing). The science club meets after school, and the members [is / **are** / was] planning a fundraiser. *(Answer: are)*

4. Text Entry / Griddable

Online: Type a number or short answer; for numbers you may fill a grid. **On paper:** Write in the box, then bubble the grid when provided.

Example (Math). A line passes through (0, 3) with slope 2. What is y when $x = 4$? [ ] *(Answer: 11)*

5. Drag-and-Drop

Online: Drag words/numbers from a bank into blanks or an order. **On paper:** A bank is printed; write each item on the correct line or draw an arrow.

Example (Math). Order from **least to greatest.** Bank: 0.4, ⅓, 0.37, ⅖ Least __ __ __ __ Greatest *(Answer: ⅓, 0.37, 0.4, ⅖)*

6. Equation Editor

Online: Build an expression/equation with an on-screen keypad. **On paper:** Write it neatly on the line.

Example (Math). Write an equation in form $y = mx + b$ for slope 3, y-intercept −5. _____ *(Answer: $y = 3x - 5$)*

7. Graphing

Online: Plot points, lines, or bars on a grid. **On paper:** Use the printed coordinate grid and a pencil.

Example (Math). On the grid provided, plot (−2, 4) and draw the line $y = x + 1$.

8. Hot Text

Online: Click a sentence or phrase in the passage. **On paper:** Underline, or write the number of, the sentence that answers the question.

Example (Reading). Underline the sentence that **best** supports the idea that the narrator felt nervous.

9. Written Responses (SCR and ECR)

Online: Type in a text box. **On paper:** Write in the lined space provided.

- **Short Constructed Response (SCR)** — a few sentences with text evidence (2 points).
- **Extended Constructed Response (ECR)** — a full multi-paragraph essay (10 points).

You will practice these in depth in the **SCR Mastery** and **ECR Mastery** sections.

> **Strategy:** When a question says "Select TWO" or "least to greatest," underline those words before you answer. Many points are lost not from wrong math or reading, but from missing the directions.

Your Study Calendars

Pick the plan that fits the time you have. Each uses the same four stages — **Learn, Practice, Simulate, Diagnose** — and spreads the eight full-length tests so you never cram. Check off each day as you go.

> **Key Idea:** Short, steady sessions beat one long rush. Aim for 30–45 focused minutes per study day.

The 4-Week Sprint (test is coming soon)

Week	Focus	Do This
1	Math — Strands 1 & 2	Read lessons; warm-up drills; take **Math Test 1**; review every explanation
2	Math — Strands 3 & 4	Lessons; drills; take **Math Tests 2 & 3**; fill diagnostic charts; redo weak skills
3	RLA — Reading + SCR	Genre guide & reading lessons; **SCR Mastery**; take **RLA Tests 1 & 2**
4	RLA — Writing + ECR, then finish	Revising/editing refresher; **ECR Mastery**; take **Math Test 4**, **RLA Tests 3 & 4**; final review

The 8-Week Plan (more time, deeper mastery)

Week	Focus	Do This
1	Math Strand 1 — Numerical	Lesson + drills; begin a daily routine
2	Math Strand 2 — Computations & Algebraic	Lesson + drills; **Math Test 1** + review
3	Math Strand 3 — Geometry & Measurement	Lesson + drills; **Math Test 2** + diagnostic chart
4	Math Strand 4 — Data & Financial Literacy	Lesson + drills; **Math Test 3**; reteach weak skills
5	RLA Reading — genres & strategies	Genre guide + reading lessons; hot-text practice

6	RLA Short Constructed Response	**SCR Mastery**; **RLA Test 1** + review
7	RLA Revising & Editing	Conventions refresher; **RLA Test 2**; **Math Test 4**
8	RLA Extended Constructed Response + final push	**ECR Mastery**; **RLA Tests 3 & 4**; final diagnostic review

Watch Out: Do not skip the review step. The biggest score gains come from understanding **why** you missed a question, then practicing that exact skill again.

Test-Day Game Plan

You have practiced the skills and the question types. Test day is simply one more practice test — this time for real. Follow this plan to stay calm and finish strong.

The Night Before

- Pack what you need and go to bed on time. Rest helps your brain more than last-minute cramming.
- Do a light review of your **diagnostic charts** only — no new material.

The Morning Of

- Eat a good breakfast. Arrive early so you feel settled, not rushed.
- Take a few slow breaths before you begin. You are ready.

During the Test — Your Pacing Plan

1. **Read the directions of each question carefully.** Watch for *select two*, *least to greatest*, and *best*.
2. **Answer the questions you know first.** Flag harder ones and return — don't let one question eat your time.
3. **Use your tools.** On Math, use the **calculator** and **reference sheet** and show work on scratch paper. On Reading, use the online **dictionary** when a word blocks understanding.
4. **Eliminate wrong answers.** Crossing out two choices doubles your odds on a tricky question.
5. **Never leave a question blank.** There is no penalty for guessing. Make your best choice and flag it to revisit.

Budgeting Time for Written Responses (RLA)

- Keep **SCR** answers short and direct — make your point, give **text evidence**, done.
- Save time for the **ECR** essay (10 points). A quick plan — thesis, two or three supporting points with evidence, a conclusion — keeps you organized. A partial, organized essay still earns points; a blank one earns none.

Before You Submit

- Return to every flagged question; make sure each has an answer.

- Check that multiselect questions have the **right number** of choices marked.
- Reread written responses once for clear sentences, correct spelling, and punctuation.

Key Idea: Finish every question with *some* answer, give the written responses real effort, and use your full time. Steady and complete beats fast and blank.

You've got this. Now turn the page and start building your skills.

Part 1 · Mathematics

How Grade 8 Math Is Built

The Grade 8 Mathematics test has **40 questions worth 48 points.** Every question is drawn from one of **four strands** (the state calls them reporting categories). Knowing the strands tells you exactly what to study — and this book teaches **every skill in every strand.**

Strand	What It Covers	On the Test
1 · Numerical Representations & Relationships	Real numbers, irrational numbers, scientific notation, ordering numbers	~4 questions
2 · Computations & Algebraic Relationships	Slope, linear equations, functions, one-variable equations	~15 questions (largest)
3 · Geometry & Measurement	Dilations, transformations, volume, surface area, Pythagorean theorem, angles	~13 questions
4 · Data Analysis & Personal Financial Literacy	Scatterplots, trend lines, mean absolute deviation, interest, saving	~8 questions

Readiness vs. Supporting Standards

- ★ **Readiness standard** — tested heavily every year; carries most of the points. Master these first.
- ○ **Supporting standard** — tested less often, but still fair game.

The Process Standards (8.1 A–G)

STAAR also weaves in **mathematical process standards**: applying math to real situations, choosing a strategy, using tools, and explaining your reasoning. You practice these inside every lesson and test — they are *how* you do the math, not a separate topic.

> **Key Idea:** Read each lesson, do the warm-up drills, then prove it on the practice tests. Use the **reference sheet** on the next page the whole way through — it is yours on the real test, too.

Grade 8 Mathematics Reference Sheet

On the real STAAR test, these formulas are provided for you. Learn where each one is so you can find it fast. (This replicates the key formulas from the STAAR Grade 8 Mathematics Reference Materials.)

Perimeter & Circumference	**Area**
Circle: $\mathbf{C = 2\pi r}$ or $\mathbf{C = \pi d}$	Triangle: $\mathbf{A = \frac{1}{2}bh}$ Rectangle / Parallelogram: $\mathbf{A = bh}$ Trapezoid: $\mathbf{A = \frac{1}{2}(b_1 + b_2)h}$ Circle: $\mathbf{A = \pi r^2}$
Surface Area (B = base area, P = base perimeter)	**Volume** (B = base area)
Prism — Lateral: $\mathbf{S = Ph}$ · Total: $\mathbf{S = Ph + 2B}$ Cylinder — Lateral: $\mathbf{S = 2\pi rh}$ · Total: $\mathbf{S = 2\pi rh + 2\pi r^2}$	Prism / Cylinder: $\mathbf{V = Bh}$ Cone: $\mathbf{V = \frac{1}{3}Bh}$ Sphere: $\mathbf{V = \frac{4}{3}\pi r^3}$
Algebra & Graphing	**Personal Financial Literacy**
Slope: $\mathbf{m = (y_2 - y_1) / (x_2 - x_1)}$ Slope-intercept form: $\mathbf{y = mx + b}$ Direct variation: $\mathbf{y = kx}$ Pythagorean theorem: $\mathbf{a^2 + b^2 = c^2}$	Simple interest: $\mathbf{I = Prt}$ Compound interest: $\mathbf{A = P(1 + r/n)^{nt}}$
Useful Constants & Conversions	
$\mathbf{\pi \approx 3.14}$ (or $22/7$) · Metric: 1 m = 100 cm = 1000 mm; 1 km = 1000 m · Customary: 1 ft = 12 in; 1 yd = 3 ft; 1 mi = 5,280 ft	

Strategy: You do **not** need to memorize these — but you must know they exist and how to use them. Every worked example that uses a formula shows you which one and why.

Strand 1 · Numerical Representations and Relationships

What this strand tests: how to represent, compare, and order real numbers — including irrational numbers and scientific notation.

Standards covered:

- ○ **8.2A** — Describe relationships between sets of real numbers using a visual representation.
- ○ **8.2B** — Approximate an irrational number (including π and square roots of numbers less than 225) and locate it on a number line.
- ○ **8.2C** — Convert between standard decimal notation and scientific notation.
- ★ **8.2D** — Order a set of real numbers from mathematical and real-world contexts. *(Readiness)*

8.2A · The Real Number System ○

Every number you use at Grade 8 is a **real number.** Real numbers divide into groups nested inside one another:

- **Natural:** 1, 2, 3, … **Whole:** 0, 1, 2, 3, … **Integers:** … −2, −1, 0, 1, 2, …
- **Rational:** any number written as a fraction a/b — integers, terminating decimals (0.75), repeating decimals (0.333…).
- **Irrational:** cannot be written as a simple fraction; decimals never end or repeat — such as **π** and **√2**.

> **Key Idea:** The groups are *nested* — every natural number is also whole, an integer, and rational. But rational and irrational never overlap.

Worked example. Classify √16 and √20. √16 = 4 → natural, whole, integer, **rational.** √20 ≈ 4.472… → **irrational.**

Watch Out: A square root is irrational only when the number is **not** a perfect square. √25 = 5 is rational; √26 is irrational.

Try It. Every integer is a rational number — true or false? *(True: any integer n = n/1.)*

8.2B · Approximating Irrational Numbers ○

To estimate a square root, find the two **perfect squares** it falls between.

Worked example. Approximate √50. $49 = 7^2$ and $64 = 8^2$, so √50 is between **7 and 8**; just above 49, so √50 ≈ **7.1** — mark it just right of 7 on a number line. **π ≈ 3.14**, between 3 and 4.

Watch Out: "Less than 225" means the number under the root is below 225 — roots up to about √224 ≈ 14.97.

Try It. Between which two whole numbers is √30? *(5 and 6, since 25 < 30 < 36.)*

8.2C · Scientific Notation ○

Scientific notation writes a number as $\mathbf{a \times 10^n}$, where $\mathbf{1 \le |a| < 10}$ and n is an integer. Large numbers → positive exponent; small numbers (less than 1) → negative exponent.

- 4,500,000 = $\mathbf{4.5 \times 10^6}$ · 0.00032 = $\mathbf{3.2 \times 10^{-4}}$ · 7.1×10^5 = **710,000**

Watch Out: The factor *a* must be at least 1 and less than 10. Writing 45×10^5 is wrong — fix to 4.5×10^6.

Try It. Write 0.0067 in scientific notation. *(6.7×10^{-3}.)*

8.2D · Ordering Real Numbers ★ (Readiness)

Numbers often appear in different forms at once — fractions, decimals, percents, roots, scientific notation. **Convert them all to decimals, then compare.**

Worked example. Order from least to greatest: ⅗, 0.58, √0.36, 62%.
⅗ = 0.60 · 0.58 = 0.58 · √0.36 = 0.60 · 62% = 0.62 → **0.58, then ⅗ = √0.36, then 62%.**

> **Key Idea:** One common form — usually the decimal — lets you compare very different-looking numbers fairly. A number line helps you *see* the order.

Watch Out: On a negative number line, the number farther **left** is **smaller**. −0.8 < −0.3.

Try It. Order greatest to least: 0.7, ⅔, 71%. *(71% = 0.71, 0.7, ⅔ ≈ 0.667.)*

Strand 1 Checklist

- ☐ 8.2A Classify real numbers and describe how the sets relate
- ☐ 8.2B Approximate irrational numbers and place them on a number line
- ☐ 8.2C Convert between standard and scientific notation
- ☐ 8.2D Order a mixed set of real numbers

Ready to practice? Warm-up drills for Strand 1 come after all four strand lessons.

Strand 2 · Computations and Algebraic Relationships

This is the **largest strand** on the test (about 15 of 40 questions). It is all about **lines, relationships, and equations.** Master it and you master most of the Math test.

Standards covered:

- ○ **8.4A** — Similar right triangles show slope is the same between any two points on a line.
- ★ **8.4B** — Graph proportional relationships; interpret the unit rate as the slope.
- ★ **8.4C** — Use a table or graph to find the rate of change (slope) and the y-intercept.
- ○ **8.5A** — Linear proportional situations as y = kx. ○ **8.5B** — Non-proportional as y = mx + b (b ≠ 0).
- ○ **8.5E** — Direct variation. ○ **8.5F** — Distinguish proportional from non-proportional. ○ **8.5H** — Examples of each.
- ★ **8.5G** — Identify functions from ordered pairs, tables, mappings, graphs.
- ★ **8.5I** — Write an equation y = mx + b to model a linear relationship.
- ○ **8.8A** — Write one-variable equations/inequalities with variables on both sides. ○ **8.8B** — Write a real-world problem for a given equation. ★ **8.8C** — Model and solve them.
- ○ **8.9A** — Identify a system's solution from the intersection of two graphs.

Part A · Slope and Rate of Change (8.4A ○, 8.4B ★, 8.4C ★)

Slope measures how steep a line is — how much *y* changes for each change in *x*:

$$\textbf{slope} = m = (\text{change in } y) \div (\text{change in } x) = (y_2 - y_1) / (x_2 - x_1)$$

- **8.4A — Why slope is constant.** Right triangles drawn to any two points on the same line are **similar**, so rise ÷ run is the **same everywhere** — that ratio is the slope.
- **8.4B — Proportional graphs.** A proportional relationship passes through the **origin (0, 0)**; equation **y = kx**; the **unit rate is the slope k.**
- **8.4C — From a table or graph.** Slope = (change in y) ÷ (change in x); the **y-intercept** is the y-value when **x = 0.**

Worked example (8.4C). Table: (0, 2), (1, 5), (2, 8). Slope = (5 − 2) ÷ (1 − 0) = **3**; y-intercept = **2**; equation y = 3x + 2.

Watch Out: Rise over run — count the **vertical** change first, then horizontal.

Try It. A line through (0, 0) and (4, 10) — slope? *(10 ÷ 4 = 2.5.)*

Part B · Proportional vs. Non-Proportional (8.5A/B/E/F/H ○)

	Proportional	Non-Proportional

Equation	y = kx	y = mx + b (b ≠ 0)
Graph	Line **through the origin**	Line **not** through the origin
Table	Constant ratio $y/x = k$	y/x is **not** constant

- **8.5E — Direct variation** is a proportional relationship y = kx. If y = 15 when x = 3, then k = 5, so y = 5x.
- **8.5F / 8.5H — Telling them apart.** Does the line pass through (0, 0)? Yes → proportional. A starting amount (y-intercept ≠ 0) → non-proportional.

Worked example (8.5B). A gym charges a $20 sign-up fee plus $5 per visit: **y = 5x + 20.** The $20 fee (b ≠ 0) makes it **non-proportional.**

Watch Out: A per-item rate with no starting fee is proportional; add any flat fee and it becomes non-proportional.

Try It. Is y = 7x proportional? *(Yes — passes through the origin, b = 0.)*

Part C · Functions (8.5G ★)

A **function** pairs **each input (x) with exactly one output (y).** In tables/mappings, an x-value may not repeat with two different y-values. On graphs, use the **vertical line test** — if any vertical line crosses the graph more than once, it is **not** a function.

Worked example. Is {(1, 3), (2, 5), (1, 7)} a function? **No** — input 1 pairs with both 3 and 7.

Try It. Does a non-vertical straight line pass the vertical line test? *(Yes — it is a function.)*

Part D · Writing Linear Equations (8.5I ★)

To write **y = mx + b**, find the **slope m** and the **y-intercept b.**

Worked example. A plant is 4 cm tall and grows 2 cm each week. Slope m = **2**; y-intercept b = **4**; equation **y = 2x + 4.**

Strategy: "Starting amount" is almost always **b**; "rate per unit" is almost always **m.**

Try It. A taxi charges $3 to start plus $2 per mile. Equation for cost y after x miles? *(y = 2x + 3.)*

Part E · One-Variable Equations and Inequalities (8.8A/B ○, 8.8C ★)

When the variable is on **both sides,** collect variable terms on one side and numbers on the other.

Worked example (8.8C). Solve 3x + 5 = x − 7. Subtract x → 2x + 5 = −7 Subtract 5 → 2x = −12 Divide by 2 → **x = −6.**

8.8A write such equations from a situation; **8.8B** do the reverse — describe a real situation the equation could model.

Watch Out: Whatever you do to one side, do to the **other** side — keep it balanced.

Try It. Solve 4x − 3 = 2x + 9. *(2x = 12, x = 6.)*

Part F · Systems of Two Linear Equations (8.9A ○)

Two lines graphed together **intersect** at the point that **satisfies both equations.** That point **(x, y)** is the **solution** of the system.

Worked example. y = x + 1 and y = −x + 5 cross at **(2, 3).** Check: 3 = 2 + 1 ✓ and 3 = −2 + 5 ✓.

Try It. If two lines cross at (−1, 4), what is the solution? *(x = −1, y = 4.)*

Strand 2 Checklist

- ☐ 8.4A–C Find and interpret slope, unit rate, and y-intercept
- ☐ 8.5A/B/E/F/H Tell proportional from non-proportional; use y = kx and y = mx + b
- ☐ 8.5G Identify functions (including the vertical line test)
- ☐ 8.5I Write an equation for a linear relationship
- ☐ 8.8A–C Write and solve one-variable equations with variables on both sides
- ☐ 8.9A Identify a system's solution from the intersection of two graphs

Strand 3 · Geometry and Measurement

This strand (about 13 of 40 questions) covers **dilations, transformations, angles, volume, surface area, and the Pythagorean theorem.** Keep the **reference sheet** open — most of these use a formula.

Standards covered:

- ○ **8.3A** — Ratio of corresponding sides of similar shapes is proportional. ○ **8.3B** — Compare a shape and its dilation on a plane. ★ **8.3C** — Algebraic rule for a scale factor (origin center).
- ○ **8.6A** — Volume formula V = Bh of a cylinder. ○ **8.6C** — Explain the Pythagorean theorem with models.
- ★ **8.7A** — Volume of cylinders, cones, spheres. ★ **8.7B** — Surface area of prisms and cylinders. ★ **8.7C** — Pythagorean theorem and its converse. ○ **8.7D** — Distance between two points.
- ○ **8.8D** — Angle sums, exterior angles, transversal angles, AA similarity.
- ○ **8.10A** — Orientation/congruence of transformations. ○ **8.10B** — Which preserve congruence. ★ **8.10C** — Rules for translations, reflections, rotations (90/180/270/360). ○ **8.10D** — Effect of a dilation on length and area.

Part A · Dilations and Similarity (8.3A ○, 8.3B ○, 8.3C ★, 8.10D ○)

A **dilation** resizes a figure by a **scale factor k** from a center. With the **origin** as center: **(x, y) → (kx, ky).**

- **8.3A / 8.3B — Similar shapes.** A figure and its dilation are **similar**: corresponding sides are proportional (×k) and angles stay the **same.**
- **8.10D — Effect on measurements.** Lengths multiply by **k**; **area** multiplies by $\mathbf{k^2}$**.**

Worked example (8.3C). Vertex (2, 3) dilated by factor 3 about the origin → (3·2, 3·3) = **(6, 9).**

Watch Out: k > 1 enlarges; 0 < k < 1 shrinks. Area changes by the **square** of k — factor 2 makes area **4×**.

Try It. A square of side 5 is dilated by k = 2 — new side? *(10.)*

Part B · Transformations on the Coordinate Plane (8.10A/B ○, 8.10C ★)

Transformation	Algebraic rule	Changes size?
Translation (slide)	(x, y) → (x + a, y + b)	No — congruent
Reflection over x-axis	(x, y) → (x, −y)	No — congruent
Reflection over y-axis	(x, y) → (−x, y)	No — congruent
Rotation 180° about origin	(x, y) → (−x, −y)	No — congruent
Rotation 90° counterclockwise	(x, y) → (−y, x)	No — congruent

Dilation (factor k)	(x, y) → (kx, ky)	**Yes** — similar, not congruent

8.10A / 8.10B — Congruence. Translations, reflections, and rotations are **rigid** — the image is **congruent.** A dilation (k ≠ 1) changes size, so it is **not** congruent, only **similar.**

Worked example (8.10C). Reflect (4, −2) over the x-axis → **(4, 2).**

Watch Out: Reflecting over the **x-axis** flips the **y**-sign; over the **y-axis** flips the **x**-sign.

Try It. Rotate (3, 5) by 180° about the origin. *((−3, −5).)*

Part C · Angle Relationships (8.8D ○)

- **Triangle angle sum:** interior angles add to **180°.**
- **Exterior angle** = sum of the two remote interior angles.
- **Parallel lines + transversal:** corresponding angles equal; alternate interior angles equal.
- **Angle-Angle (AA) similarity:** two equal angles → triangles are **similar.**

Worked example. Angles 50° and 60° → third = 180 − 50 − 60 = **70°.**

Try It. Angles 90° and 35° — find the third. *(55°.)*

Part D · Volume (8.6A ○, 8.7A ★)

- **Cylinder:** V = Bh = $\pi r^2 h$. **Cone:** V = $\frac{1}{3}\pi r^2 h$ (one-third of the cylinder). **Sphere:** V = $\frac{4}{3}\pi r^3$.

Worked example (8.7A). Cylinder radius 3, height 10: V = $\pi(3^2)(10)$ = **$90\pi \approx 282.6$** cubic units.

Watch Out: The cone is ⅓ of the matching cylinder — don't forget the one-third.

Try It. Cone radius 3, height 10 — volume? *(⅓·90π = 30π ≈ 94.2.)*

Part E · Surface Area (8.7B ★)

- **Prism — total:** S = Ph + 2B. **Cylinder — total:** S = $2\pi rh + 2\pi r^2$ (side plus two circles).

Worked example (8.7B). Cylinder radius 2, height 5: S = $2\pi(2)(5) + 2\pi(2^2) = 20\pi + 8\pi$ = **$28\pi \approx 87.9$** square units.

Strategy: "Lateral" = the **side** only; "total" **adds the bases.** Read which one is asked.

Try It. Lateral surface area of that cylinder? *($2\pi rh = 20\pi \approx 62.8$.)*

Part F · The Pythagorean Theorem (8.6C ○, 8.7C ★, 8.7D ○)

For a **right triangle** with legs a and b and hypotenuse c: $\mathbf{a^2 + b^2 = c^2}$.

- **8.7C** — Find a missing side, or use the **converse** to test whether a triangle is right.

- **8.7D** — Distance between two points: horizontal and vertical gaps are the legs; the distance is the hypotenuse.

Worked example (8.7C). Legs 6 and 8: $c^2 = 36 + 64 = 100 \rightarrow c =$ **10.** **(8.7D)** Distance (1, 2) to (4, 6): legs 3 and 4 $\rightarrow \sqrt{(9 + 16)} =$ **5.**

Watch Out: The hypotenuse **c** is across from the right angle and is the **longest** side.

Try It. Is 5, 12, 13 a right triangle? *(Yes — 25 + 144 = 169 = 13^2.)*

Strand 3 Checklist

- ☐ 8.3A–C, 8.10D Dilate figures; use scale factor; length ×k and area ×k^2
- ☐ 8.10A–C Apply translation, reflection, rotation rules; know what stays congruent
- ☐ 8.8D Use angle sums, exterior angles, transversals, AA similarity
- ☐ 8.6A, 8.7A Volume of cylinders, cones, spheres
- ☐ 8.7B Lateral and total surface area of prisms and cylinders
- ☐ 8.6C, 8.7C, 8.7D Pythagorean theorem, converse, and distance